Pantea Mohamadkhani

Data Science & AI | Business Intelligence & Analytics
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PROFESSIONAL SUMMARY

Data Analyst with an M.Sc. in Artificial Intelligence & Robotics, a B.Sc. in Industrial Engineering, and a completed MBA. Core strengths: AI, SQL, Git, NumPy, Python, Pandas. Hands-on data professional who turns raw, messy data into structured pipelines, statistical analysis, and clear dashboards. Portfolio spans applied machine learning, NLP, and business intelligence, focused on delivering decision-ready insights to business stakeholders.

SKILLS

Languages: Python, SQL
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
BI & Analytics: Power BI, Power Query, DAX, Data Modeling, Star Schema, Dashboard Development, Data Visualization, Excel, Word, PowerPoint
Databases: SQL Server (T-SQL), Snowflake, OLAP Databases, Querying, Joins, Aggregations, Subqueries, CTEs, Window Functions
Data Engineering: ETL, Apache Spark, Databricks, Data Warehouse, Data Pipelines
Data Governance: Data Quality, Metadata Management, Data Lineage, Access Control
Version Control: Git, GitHub, Gitlab
REST API Fundamentals: HTTP methods, endpoints, JSON, request/response concepts, Python Requests, JSON data handling
Data & ML: EDA, Data Cleaning, Data Transformation, Feature Engineering, Statistical Analysis, Machine Learning, Classification, Regression, Model Evaluation, Clustering, Unsupervised Learning, NLP & Embeddings, Recommender Systems
Business Analytics: KPI Analysis, Sales Analysis, Trend Analysis, Customer Segmentation, Sales & Customer Behavior Analysis, Management Reporting
Other: Reinforcement Learning, Prompt Engineering, Business Model Design

PROFESSIONAL EXPERIENCE

Data Analyst
2023 - 2025

Adrin Darou Iranian - Karaj, Iran

- Built and maintained Excel and Power BI reports and dashboards that gave management recurring visibility into sales performance, product profitability, and top- and low-performing items.
- Analyzed sales and pricing data for pharmaceutical, supplement, and medical-equipment product lines, identifying demand patterns and margin trends to support pricing and purchasing decisions.
- Cleaned, consolidated, and validated transaction, product, and customer data into structured datasets, improving reporting reliability and reducing manual spreadsheet effort.
- Evaluated product profitability and price competitiveness across categories, highlighting high-margin and underperforming items to guide commercial decisions.
- Presented analytical findings and recommendations to non-technical management, translating sales data into practical pricing and sales actions.

Data Analyst Intern
Mar 2022 - Jun 2022

MAPNA Turbine Engineering & Manufacturing Co. (TUGA) - Alborz Industrial City, Iran

- Supported analytical workflows within an industrial turbine manufacturing facility, processing production and operational datasets.
- Implemented standardized data-cleaning procedures and reporting practices to improve business-oriented operational visibility.
- Collaborated with cross-functional teams to streamline data preparation pipelines for manufacturing analytics.

PROJECTS

E-commerce Orders Lens - Data Cleaning, EDA & Machine Learning

Data Analyst & ML Developer - Python, Pandas, NumPy, scikit-learn, Matplotlib, Jupyter, KMeans

https://github.com/panteamkhh/ecommerce-orders-lens

- Ran a documented 21-step cleaning pipeline on a synthetic e-commerce orders dataset to produce an auditable, analysis-ready dataset.
- Performed question-by-question exploratory data analysis (EDA) to surface patterns in orders, customers, and revenue.
- Trained and evaluated machine-learning models with scikit-learn, including KMeans customer segmentation and a classification task.

- Built a self-contained HTML dashboard, covered by 34 automated tests and CI.

PharmaSales360 -Pharmaceutical Sales & CRM Analytics

Data / BI Analyst - Python, Pandas, SQL Server, T-SQL, Power BI, DAX, Star Schema, RFM Segmentation

https://github.com/panteamkhh/pharma-sales360

- Built an end-to-end pharma sales and CRM analytics project: reproducible data generation, SQL Server star schema, statistical analysis, and Power BI reporting.
- Designed a star-schema model in SQL Server and wrote T-SQL queries for sales, customer, and product analysis.
- Performed statistical analysis and RFM customer segmentation to identify high-value customers and revenue drivers.
- Specified Power BI dashboards with DAX measures and translated results into concrete business recommendations.

Insurance Data Analysis

Data Analyst - Python, Pandas, NumPy, Matplotlib, Seaborn

https://github.com/panteamkhh/insurance-data-analysis

- Analyzed a synthetic portfolio of 10,000 policies across Travel, Health, Auto, Life, and Home in Python.
- Answered core business questions â€” book size, claims behavior, loss-making lines, and premium vs cover â€” via reusable, tested functions.

CEFR Lexical Intelligence -End-to-End NLP System

Lead Developer & Researcher - Python, Pandas, scikit-learn, NLP Embeddings, Matplotlib

https://github.com/panteamkhh/cefr-lexical-intelligence

- Designed and curated a custom dataset of 14,848 vocabulary entries classified across CEFR levels (A1-C2) and semantic domains.
- Built an end-to-end 9-stage ML/NLP pipeline covering data cleaning, EDA, classification, embeddings, unsupervised clustering, and semantic search.
- Implemented sentence embeddings and clustering to detect latent lexical groupings beyond predefined CEFR labels.

Sales Data Analysis (Python + Power BI)

Data Analyst - Python, Pandas, Power BI, DAX, Excel

https://github.com/panteamkhh/sales-data-analysis

- Analyzed a retail sales dataset (3,510 orders with customers, products, and promotions) question by question in Python.
- Packaged the results into an interactive Power BI dashboard so business users could explore sales, profit, and geographic trends without code.

Maintenance Data Pipeline - End-to-End CMMS Analytics

Data Analyst & Pipeline Developer - Python, Pandas, Streamlit, Pytest, GitHub Actions, Root Cause Analysis

https://github.com/panteamkhh/maintenance-data-pipeline

- Built an end-to-end maintenance (CMMS) analytics pipeline: data-quality audit, cleaning, KPI calculation, Pareto/root-cause analysis, and an interactive dashboard.
- Defined and computed maintenance KPIs â€” MTTR, MDT, MWT, availability, PM compliance, and breakdown ratio â€” with documented formulas.
- Conducted process analysis of maintenance workflows to identify bottlenecks and continuous-improvement opportunities.
- Delivered a Streamlit KPI dashboard and automated management-summary reports for stakeholders, backed by tests and CI.

PUBLICATIONS

Resolving Data Sparsity in Recommender Systems Using Reinforcement Learning 2026

Accepted - Babol Noshirvani University of Technology

TEACHING & TRAINING

Data Storytelling: From Data to Effective Narrative

Instructor - Maktabkhooneh

maktabkhooneh.org/course/mk16581

EDUCATION

Master of Business Administration (MBA), Business Administration 2025 - 2026

University of Tehran

Master of Science, Artificial Intelligence & Robotics 2022 - 2025

Kharazmi University

Thesis: Addressing Data Sparsity in Recommender Systems via Reinforcement Learning

Bachelor of Science, Industrial Engineering 2017 - 2021

Islamic Azad University

CERTIFICATIONS

Generative AI for Beginners

Microsoft - 2026

Prompt Engineering for Developers

DeepLearning.AI - 2026

Complete Data Analyst Bootcamp (62h)

Udemy - 2025

Machine Learning

Andrew Ng, Coursera – 2023

Data Visualization with Python Masterclass | Python A-Z

Udemy -2022

LANGUAGES

English (Upper-Intermediate), Spanish (Basic)